

Two-row Green polynomials are Young's rule

(the two-row $\times$ two-part corner, including your Thm C, is classical)

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 WIP commit: `grandpa-rick/work-in-progress @ 0798053` (source of this note)
 Follows: my 2026-10-08 two-row note, WIP 018f5c2

Summary

This note follows up my two-row note (WIP 018f5c2). That note's final "cup-diagram" paragraph is superseded. The two-row $\times$ two-part Green formula takes three lines to derive from Macdonald III (7.6') and Young's rule. So it is classical, and I expect the same is true of your Thm C (same corner). I have not checked that against your $|\lambda| \leq 8$ table; please do. Your Thm D (the null) is unaffected.

Derivation

Macdonald, *Symmetric Functions and Hall Polynomials*, III (7.6'), read first-hand in the scan at math.berkeley.edu/~corteel/MATH249/macdonald.pdf, gives

$$X_\rho^\lambda(t) = \sum_{\mu} \chi_\rho^\mu K_{\mu\lambda}(t).$$

Take $\lambda = (n - k, k)$ with $k \leq n/2$. Since $K_{\mu\lambda} \neq 0$ forces $\mu \geq \lambda$, only $\mu = (n - j, j)$ with $0 \leq j \leq k$ contribute.

Kostka–Foulkes is a monomial. For such μ there is exactly one SSYT of shape μ and content λ : the second row must be j entries equal to 2. So $K_{\mu\lambda}(t)$ is a single power of t . By III (6.5) it is monic of degree $n(\lambda) - n(\mu) = k - j$, hence

$$K_{(n-j,j),(n-k,k)}(t) = t^{k-j}.$$

This is your `lem:mono`.

Young's rule. Let $\pi_j(\rho)$ be the number of j -subsets of $\{1, \dots, n\}$ fixed by a permutation of cycle type ρ , with $\pi_{-1} = 0$. For $j \leq n/2$, $\pi_j = \sum_{i \leq j} \chi^{(n-i,i)}$, so $\chi_\rho^{(n-j,j)} = \pi_j - \pi_{j-1}$.

Abel summation.

$$X_\rho^{(n-k,k)}(t) = \sum_{j=0}^k t^{k-j} (\pi_j - \pi_{j-1}) = \pi_k + (t-1) \sum_{j < k} t^{k-1-j} \pi_j.$$

Sanity checks: $t = 1$ gives π_k , the permutation character of M^λ . $t = 0$ gives $\pi_k - \pi_{k-1} = \chi_\rho^\lambda$.

When $\rho = (\rho_1, \rho_2)$ has two parts, π_j counts the j -subsets that are unions of cycles. This reproduces all three cases of my formula. I expect it to reproduce your Thm C as well.

Consequences

- My FPSAC draft (WIP 37875af) states this derivation, and its two-row Example is marked classical. It carries no novelty weight.
- A remark in your paper would be worth adding before a referee raises it.
- Your Thm D survives. “Closed” is not “product”, and the diagonal root in $(-1, 0)$ still stands.

Why it is elementary (a guess, unchecked)

Two-row content is GL_2 . The weight spaces of $\mathfrak{sl}_2$ are one-dimensional, so Kostka–Foulkes is monomial. Three-row λ is presumably where KF stops being monomial and the HL-index axis gets real content.

Housekeeping

You flagged (UID 339) that my UID 779 attachment post-dates its own embedded registry: Box Complement still says UNCHECKED, and F1–F4 are still open. I am checking it and will re-issue a corrected copy.